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St. Joseph's College of Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION-APRIL 2018

BC507S-MOLECULAR BIOLOGY

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL questions:

I choose the correct answer: (10X1=10)

1. The proteins required for DNA packaging is

   a) Histones b) Albumin c) Globulin d) Prothrombin.

2. Transcriptionally active portion of gene in the chromosome is

   a) Heterochromatin b) Euchromatin c) satellite DNA d) Telomere.

3. Replication of DNA is

   a) Conservative b) Semiconservative

   c) Dispersive d) None of the above.

4. Okasaki fragments are made up of

   a) RNA b) DNA c) DNA & RNA d) RNA & proteins.

5. The structure of tRNA appears like

   a) Helix b) Hairpin c) Clover leaf d) Coil.

6. Direction of RNA synthesis is

   a) 5' → 3' b) 3' → 5' c) both A & B d) none of the above.

7. Streptomycin is an inhibitor of

   a) Replication b) Transcription c) Translation d) All of the above.

8. Genetic code flows from

a) DNA to DNA b) RNA to DNA

c) RNA to proteins d) DNA to RNA.

9. DNA damage is caused by

a)Chemicals b) Radiation

c) Environmental factors d) All of the above.

10. The final repair mechanism of damaged DNA is

a) SOS repair b) Excision repair

c) Mismatch repair d) All of the above.

II. True or False (5X1=5)

11. RNA is the genetic material for all eukaryotes.
12. Synthesis of DNA is called DNA replication.
13. σ –factor is involved in termination of transcription.
14. Non-sense codons are present in mRNA.
15. Photoreactivation repair cleaves Thymine dimer.

III. Answer in one sentence (10X1=10)

16. What is cot value?
17. What is telomere?
18. What is the role of helicase in DNA replication?
19. What is okasaki fragment?
20. What do you mean by polycistronic mRNA?
21. Define Transcription.
22. What is the role of sigma factor?
23. Define genetic code.

24. What is mismatch repair?
25. What is SOS repair?

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Write a short note on organization of chromosome.
27. Write a note on DNA polymerase of prokaryotes.
28. Discuss the termination of transcription process in prokaryotes.
29. Specify the site of action of the inhibitors of protein synthesis.
30. Explain the mechanism of DNA repair by photoreactivation.
31. Write a note on lac operon.

SECTION-C (3X10=30)

Answer any THREE of the following:

32. Explain the experimental evidence to prove that the DNA is the genetic material.
33. Describe the events of replication in prokaryotes.
34. Explain the mechanism of excision and Recombination repair.
35. Explain the post transcriptional modification of mRNA.
36. Explain the features of genetic code.

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